1 Adaption of 7D-DVF

The extraction framework asks for a grade from 1 to 5, with a clear range. In this section, the intermediate values of those ranges are defined, to avoid subjectivity and bias.

1.1 Researcher autonomy

This dimension refers to the degree of freedom granted to users in interacting with shared data. This spans the following range:

- **5 - Very high autonomy**: allows full custom code execution, algorithm development, and exploratory querying.
- **4 - High autonomy**: custom code execution and analyses, but within a constrained environment with predefined security controls.
- **3 - Moderate autonomy**: restricts users to pre-approved scripts, tools, or interfaces but allows flexibility in selecting methods
- **2 - Low autonomy**: choice from a limited set of analyses or workflows.
- **1 - Very low autonomy**: limits interactions to fixed queries or predefined outputs, with no ability to modify procedures.

Higher researcher autonomy can accelerate discovery, but increases risk of unauthorized re-identification.

1.2 Data distribution

This dimension refers to the physical or virtual infrastructure in which shared data reside during visiting.

- **5 - Very high distribution**: Fully decentralized models, where analytic agents operate at the data source without relocating it.
- **4 - High distribution**: Distributed systems with unified interfaces
- **3 - Moderate distribution**: Data is stored across several institutions, but limited federation.
- **2 - Low distribution**: Single institution, several servers
- **1 - Very low distribution**: In a single database, repository or cloud environment.

Data location directly shapes who controls data, under which jurisdiction, and subject to which norms of sovereignty, accountability, and trust.

1.3 Data visibility

Data visibility refers to the extent to which users can view or explore the underlying data during a data visit.

- **5 - Very high visibility**: Full visibility grants access to datasets with raw data.
- **4 - High visibility**: Access to most individual-level data, with some variables or records restricted.
- **3 - Moderate visibility**: Partial visibility restricts access to metadata, sample records, previews, or aggregate subsets.
- **2 - Low visibility**: Access restricted to statistical summaries, visualizations or aggregates.
- **1 - Very low visibility**: Restricted visibility or query-only access allows users to submit queries to remote datasets without ever seeing the underlying data.

Visibility configurations directly shape research utility and privacy risk. Full visibility can enhance interpretability and accelarate muli-omics or AI-powered genomic reserach, especially when complex pattern detection or predictive model development is required. However, it also increases exposure and re-identification risk.

1.4 Data identifiability

This dimension refers to the type and sensitivity of information shared during visiting.

- **5 - Very high identifiability**: Personal data, including direct identifiers
- **4 - High identifiability**: Pseudonymized data, coded and re-identifiable only with a separate key
- **3 - Moderate identifiability**: De-identified data, identifiers removed, re-identification possible through linkage
- **2 - Low identifiability**: Anonymized data, irrevocably stripped of identifiers, rendering re-identification infeasible.
- **1 - Very low identifiability** Synthetic data

The nature of the shared date directly impacts the regulations it must comply with.

1.5 Output governance

This dimension refers to the controls governing the release and use of results generated through data visiting.

- **5 - Very high governance**: Pre-approved formats that restrict dissemination to structured, anonymized templates.
- **4 - High governance** Output after privacy-enhanced mechanisms such as differential privacy via noise injection or result rounding
- **3 - Moderate governance**: Systematic review before release, involving manual or automated vetting such as re-identification risk checks or cell size suppression
- **2 - Low governance**: Outputs undergo limited review, focused on high-risk disclosures, involving manual or automated vetting such as re-identification risk checks or cell size suppression
- **1 - Very low governance**: Unrestricted export of findings

1.6 Trust and control distribution

This dimension refers to how governance is structured, how accountability and oversight are distributed.

- **5 - Very high distribution**: Fully distributed models spread responsibility across a network of peers governed by shared standards
- **4 - High distribution** Distributed agents with control, trust in mechanisms embedded computationally, in secure agents or environments that enforce constraints automatically.
- **3 - Moderate distribution**: Governance responsibilities are shared between data providers and a coordinating authority.
- **2 - Low distribution**: Localized models where oversight is conducted by the institution hosting the data
- **1 - Very low distribution**: Fully centralized with a single authority or platform to administer access and monitor compliance

1.7 Auditability and traceability

This dimension refers to the capacity to monitor and record user activities and data interactions during visiting.

- **5 - Very high auditability**: Full auditability, comprehensive real-time logs of all actions enabling retrospective review
- **4 - High auditability**: Embedded traceability, privacy-preserving auditing built into computational workflows, such as automated provenance tracking.

- **3 - Moderate auditability**: Structured traceability but not automated in everything.
- **2 - Low auditability**: Limited monitoring, periodic checks or partial logs focused on key events
- **1 - Very low auditability**: No logging in the system, up to individual entities.